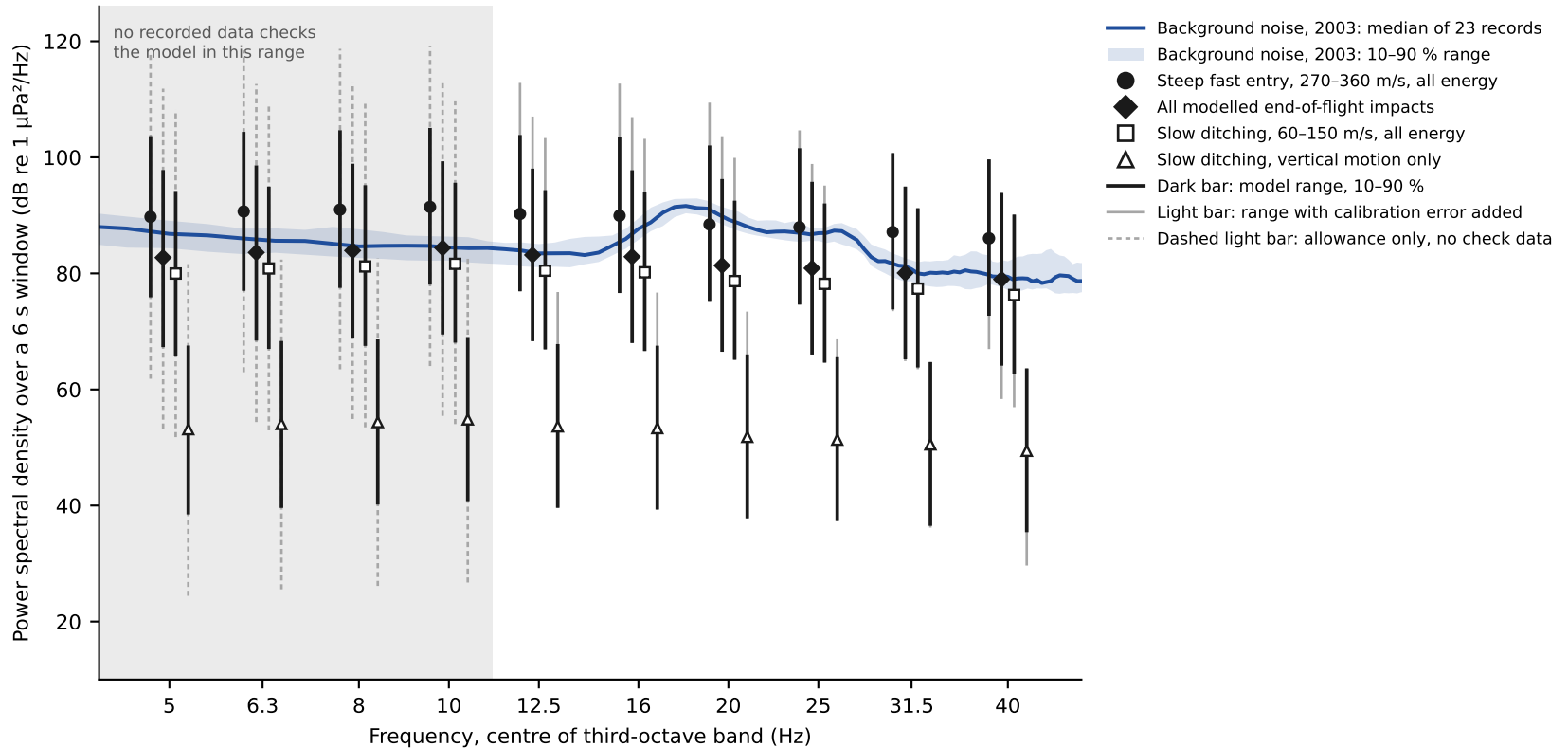


Cape Leeuwin hydrophone station (off south-west Australia): background noise and the sound level an impact would make, with the calibration error added



What the chart shows. The curve is the background noise at Cape Leeuwin. The markers are the sound level that an impact near 37.2°S 89.6°E (2,221 km away) would make at the station, one set for each frequency band. The dark bars are the range from the model. The light bars add the error that we found when we compared the model with airgun shots recorded in 2001. Below 12.5 Hz, no recorded data checks the model, so the dashed bars there are an allowance of ± 14 dB, not a measurement. If a marker is above the curve, the impact can be heard in that band. These are estimates. They are not a probability of detection.

Technical. Received ESD = $\eta_{\text{cal}}(\tau, z_s) \cdot E_{\text{pc}} / 2\pi \cdot S_b(\tau) \cdot 10^{-(\text{TL}_b/10)} \cdot 10^{-(C/10)}$, shown as PSD over 6 s per third-octave band; η_{cal} from the F-35A (H11, 0.7 Pa, 900 MJ, RAM-corrected), $\tau \sim \log\text{-U}(0.05, 10)$ s, $z_s \in \{2, 10, 30\}$ m, $\text{ESD} \propto F^2$ above $1/\tau$, $C \sim N(0, 10) + N(0, 3)$ dB; KRAKEN adiabatic incoherent modes, hard half-space, median impact path minus RAM Δ ; $m = 175$ t; $n = 40,000$ draws per scenario (module arithmetic reproduced to 0.0 dB). Scenario 'all modelled' = core run (b) end-of-flight impacts, 00:19 option R600 BTO only, prior weights; 'steep fast entry' tops out above the sea-level speed of sound. Calibration error: $\text{RL}_{\text{true}} = \text{RL}_{\text{model}} - r_b$, $r_b = \text{observed} - \text{model TL}$ for air9 at this station (Blackman et al. 2004, Fig. 23, p. 21; 10 m source; dB: 12.5 Hz -9, 16 Hz -9, 20 Hz -7, 25 Hz -3, 31.5 Hz +0, 40 Hz +6, 50 Hz +8, 63 Hz +14); outside that span $\pm \max|r_b|$. Audit verdict for air9: level PASS, slope FAIL at both stations (not calibrated in slope). Noise: Blackman et al. 2004, App. B, pp. 25-31, 6 s pre-event windows, counts $\rightarrow$ Pa by the FDSN response with SAC scaling inferred, not stated (route R); May-June 2003, not March 2014.